

# CAPSTONE PROJECT

## ATLANTIC CANADA CRIME PATTERNS ANALYSIS: CLUSTERING LOCALITIES BY TYPE, STRUCTURE, AND GROWTH (2020–2024)

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# PROBLEM DEFINITION

Identify patterns of criminal activity within Atlantic Canadian provinces based on similarities in crime trends between 2020 and 2024.

Provinces:

- Nova Scotia
- New Brunswick
- Newfoundland and Labrador
- Prince Edward Island

## Main Question:

Can we group localities based on similar crime patterns?



# WHY IS THIS IMPORTANT?

Raw numbers do not always show patterns clearly.

- ✓ Two provinces may have different total crime levels
- ✓ But proportions of crime types may be similar
- ✓ Or growth trends may show similar patterns over time



# VALUE & BENEFICIARIES

- ✓ **Policymakers** allocate resources efficiently
- ✓ **Researchers** conduct comparative studies
- ✓ **Public** better understand crime trends
- ✓ Identify locations needing specialized strategies

# WORKFLOW OVERVIEW 1/2

**1. Data Processing and Exploration**



**2. Feature Engineering & Normalization**



**3. Unsupervised Machine Learning (Clustering)**



**4. Interpretation and Reporting**

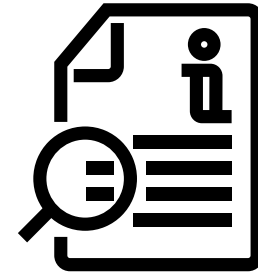


# WORKFLOW OVERVIEW 2/2



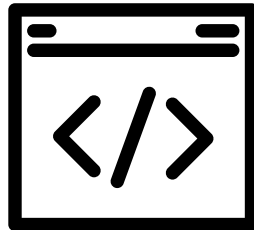
## 1. Data Processing Techniques

- Cleaned raw CSVs
- Removed headers/footers
- Handled missing values



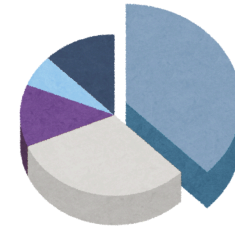
## 2. Feature Engineering

- Transform counts → proportions
- Scale features
- Prepare for clustering



## 3. Machine Learning Technique

- No predefined labels
- Cluster localities by similarity
- Silhouette score → optimal clusters
- PCA → visualize clusters



## 4. Interpretation and Reporting

- Understanding clusters
- Analyze cluster centroids
- Identify localities with distinct patterns

# TOOLS AND LIBRARIES



- **Pandas:** Data manipulation and cleaning
- **NumPy:** Numerical operations
- **Matplotlib:** Adjustments and custom plot details.
- **Seaborn:** Main visualizations for style and ease.
- **Scikit-learn:** Machine learning (K-means, PCA, scaling)

# WHY DBSCAN WAS NOT USED?

- DBSCAN works best with density-based clusters
- Most localities have similar crime structures
- DBSCAN could group most points together or mark many as noise
- K-Means fits better for grouping localities by overall similarity



# CHALLENGES & LIMITATIONS

- ✓ Most localities are structurally similar → clusters concentrate.
- ✓ Only aggregated crime counts → no socioeconomic/demographic context
- ✓ Short time period (2020–2024) → limited historical perspective
- ✓ Large cities may dominate totals without scaling

# ASSUMPTIONS

- ✓ Feature scaling and proportional transformation represent meaningful structure
- ✓ Crime categories used are sufficient to compare patterns
- ✓ Clustering can reveal meaningful groups despite lack of labels

# CONCLUSIONS

- ✓ Data preparation is critical
- ✓ Proportions can be more meaningful than totals
- ✓ Machine learning does not replace human interpretation
- ✓ Generative AI helped structure code and documentation

**THANK YOU**

